

01-08-2016

Data - Flow Integrity

Initial Assumptions

Software: C++ , LLVM Pass?

// Note : Compiler's as a subject is scheduled for 6th Semester but I have some basic ideas due to Automata..

Target / Goals:

- Count memory load/store instructions
- Define a module?
 - Maybe a function?
 - Maybe a class?
 - Abstract Syntax Tree of program?
- Small C program and a Script (Makefile?)

Research

LLVM

Source code $\rightarrow$ AST $\rightarrow$ IR $\rightarrow$ Optimised IR $\rightarrow$ Machine Code
Multiple Passes (LLVM Pass)

Will I have to look into ^{LLVM} ~~gcc~~ IR ? The compilation process?

#include "llvm stuff"

Ok that is simpler that we are provided access by import.

What is a module in this context?

Could not find anything relevant...

Chotgpt:

- 1) LLVM gives the pass a Module Object
- 2) The module contains all functions in the compiled code
- 3) In context of LLVM module is the top level container that represents all functions in compiled code

- Compile Program into LLVM IR
- Run my pass
- Profit??

How to find instances of load/store in IR? (Read how IR is represented.)

LLVM IR separates memory from registers.

$\%r1 = \text{load } \frac{c32}{c64}, \text{ ptr } \%r2$

Store $\frac{c32}{c64} \text{ } \%r1, \text{ ptr } \%r2$

Special Case: atomic operations

atomicrll

02-08-2026

01-05
What is LLVM?

LLVM The compiler infrastructure

CLANG C, C++ and Object C frontend built
On top of LLVM

LLVM IR Intermediate Representation for optimisation

LLVM maintains a rich program representation
throughout software lifecycle.

Separated Frontend, Optimisation and Backend

LLVM allows program optimisation at compile, link, run
and idle time.

SSA: Static Single Assignment is a type of IR
in which each variable is assigned exactly once.

Infinite Virtual Registers